1.6 DEGREE VECTORS OF SIMPLE GRAPHS

The **degree vector** $d(G)$ of a simple graph G is the sequence of degrees of its vertices arranged in nonincreasing order. If G and G' are isomorphic, $d(G) = d(G')$. But the converse is not true. The three graphs in Fig. 1-3 have all the same degree vector [3 3 3 3 3 3], but G_3 is not isomorphic to G_1 or G_2.

Every simple graph has a unique degree vector (which can be easily constructed), and the sum of the terms in the vector is even. Notice that each term in a degree vector with n components is nonnegative and is at most $(n - 1)$, where n is the order of the graph. On the other hand, a finite nonincreasing vector $v = [d_1 \quad d_2 \quad \cdots \quad d_n]$ of nonnegative integers, where each $d_i \leq (n - 1)$ and the sum of the terms is even, need not be the degree vector of a simple graph. Consider, for example, the vector $v = [3 \quad 3 \quad 3 \quad 1]$. If there were a simple graph with v as a degree vector, the subgraph obtained by deleting the vertex of degree 1 (with three vertices and four edges) would not be simple.

A vector v is called a **graphical vector** if there exists a simple graph such that v is the degree vector of that graph. Thus [3 3 3 1] is not a graphical vector. The following theorem gives a necessary and sufficient condition for a vector to be a graphical vector.

Theorem 1.6 **(Hakimi–Havel).** Let $v = [d_1 \quad d_2 \quad d_3 \quad \cdots \quad d_k]$ be a nonincreasing vector of k (where k is at least 2) nonnegative integers such that no component d_i exceeds $(k - 1)$. Let v' be the vector obtained from v by deleting d_1 and subtracting 1 from each of the next d_1 components of v. Let v_1 be the nonincreasing vector obtained from v' by rearranging its components if necessary. Then v is a graphical vector if and only if v_1 is a graphical vector. (See Solved Problem 1.68.)

Example 8. Consider $v = [5 \quad 4 \quad 3 \quad 3 \quad 3 \quad 3 \quad 3 \quad 2]$ with eight components in which no component exceeds 7.

Delete the first component 5, and subtract 1 from the next five components of v. We obtain $v' = [3 \quad 2 \quad 2 \quad 2 \quad 2 \quad 3 \quad 2]$. By rearranging the components of v', we get the nonincreasing vector $v_1 = [3 \quad 3 \quad 2 \quad 2 \quad 2 \quad 2 \quad 2]$ with seven components.

According to Theorem 1.6, v is a graphical vector if and only if v_1 is a graphical vector. Proceeding further, we see in the next iteration that v_1 is a graphical vector if and only if $v_2 = [2 \quad 2 \quad 2 \quad 2 \quad 1 \quad 1]$ is a graphical vector. At the next iteration, we see that the vector v_2 is a graphical vector if and only if $v_3 = [2 \quad 1 \quad 1 \quad 1 \quad 1]$ is a graphical vector. Then v_3 is a graphical vector if and only if $v_4 = [1 \quad 1 \quad 0 \quad 0]$ is a graphical vector. At the next stage, we get $v_5 = [0 \quad 0 \quad 0]$. Now v_5, being the degree vector of a simple graph of order 3 with no edges, is a graphical vector. So the given vector v is also a graphical vector.

Algorithm to Test Whether a Given Vector Is Graphical

The input is a nonincreasing vector v with integer components.

Step 0. (Initialization) The current vector is v.

Step 1. If the current vector with k components has a component that exceeds $(k - 1)$, go to step 5. Otherwise, go to step 2.

Step 2. If the current vector has a negative component, go to step 5. Otherwise, go to step 3.

Step 3. If the current vector is the zero vector, go to step 6. Otherwise, go to step 4.

Step 4. (Iteration) Rearrange the components of the current vector so that it becomes nonincreasing with d_1 as the first component. Delete d_1 from the rearranged vector, and subtract 1 from each of the next d_1 components of the rearranged vector. The vector thus constructed is the updated current vector. Go to step 1.

Step 5. The input vector is not graphical. Go to step 7.

Step 6. The input vector is graphical. Go to step 7.

Step 7. Stop.

Example 9. Using this algorithm it can be verified that $v = [5 \quad 4 \quad 4 \quad 3 \quad 3 \quad 3 \quad 2]$ is a graphical vector.

Iteration 1:

$$v = [5 \quad 4 \quad 4 \quad 3 \quad 3 \quad 3 \quad 2] \quad \text{and} \quad v_1 = [3 \quad 3 \quad 2 \quad 2 \quad 2 \quad 2]$$

Iteration 2:

$$v = [3 \quad 3 \quad 2 \quad 2 \quad 2 \quad 2] \quad \text{and} \quad v_1 = [2 \quad 2 \quad 2 \quad 1 \quad 1]$$

Iteration 3:

$$v = [2 \quad 2 \quad 2 \quad 1 \quad 1] \quad \text{and} \quad v_1 = [1 \quad 1 \quad 1 \quad 1]$$

Iteration 4:

$$v = [1 \quad 1 \quad 1 \quad 1] \quad \text{and} \quad v_1 = [1 \quad 1 \quad 0]$$

Iteration 5:

$$v = [1 \quad 1 \quad 0] \quad \text{and} \quad v_1 = [0 \quad 0]$$

At the end of the fifth iteration we get the zero vector. So the given vector v is a graphical vector.

A simple graph for which the graphic vector v given above is the degree vector is shown in Fig. 1-4.

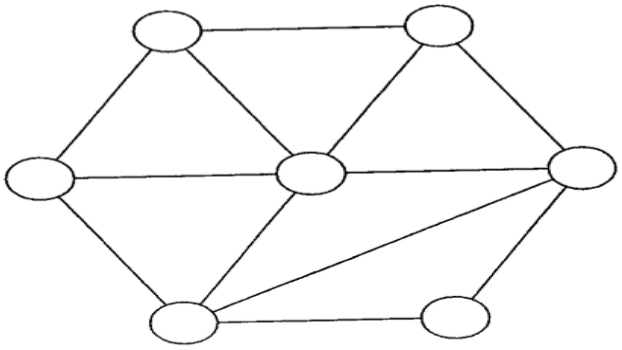


Fig. 1-4

Example 10. Using the algorithm, we now show that the vector $v = [3 \quad 3 \quad 3 \quad 1]$ is not a graphical vector.

At the end of the first iteration, we get $[2 \quad 2 \quad 0]$ as the current vector. At the end of the next iteration, we get $[1 \quad -1]$ as the current vector, which has a negative component. The conclusion is that v is not a graphical vector.

Even if a vector is graphical, it is not the case that it is the degree vector of a unique (up to isomorphism) simple graph. The vector $v = [3 \quad 3 \quad 3 \quad 3 \quad 3 \quad 3]$ is graphical, but both G_1 and G_3 in Fig. 1-3 have v as the degree vector.

1.70 Test whether $[5 \quad 4 \quad 3 \quad 3 \quad 3 \quad 3 \quad 3 \quad 2]$ is a graphical vector. If it is graphical, draw a simple graph with this vector as the degree vector.

Solution.

Iteration 1: $v = [5 \quad 4 \quad 3 \quad 3 \quad 3 \quad 3 \quad 3 \quad 2]$ and $v_1 = [3 \quad 3 \quad 2 \quad 2 \quad 2 \quad 2 \quad 2]$

Iteration 2: $v = [3 \quad 3 \quad 2 \quad 2 \quad 2 \quad 2 \quad 2]$ and $v_1 = [2 \quad 2 \quad 2 \quad 2 \quad 1 \quad 1]$

Iteration 3: $v = [2 \quad 2 \quad 2 \quad 2 \quad 1 \quad 1]$ and $v_1 = [2 \quad 1 \quad 1 \quad 1 \quad 1]$

Iteration 4: $v = [2 \quad 1 \quad 1 \quad 1 \quad 1]$ and $v_1 = [1 \quad 1 \quad 0 \quad 0]$

Iteration 5: $v = [1 \quad 1 \quad 0 \quad 0]$ and $v_1 = [0 \quad 0 \quad 0]$

So the given vector is graphical. A simple graph with v as degree vector is shown in Fig. 1-20.